

KUNUNURRA, AustraliaWMO: **942160**Lat: **15.7814S**Lon: **128.7100E**Elev: **146**StdP: **14.62**Time Zone: **8.00 (E08)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	51.9	54.8	14.1	11.3	83.0	19.4	14.7	81.5	18.8	74.2	17.6	75.4	2.0	130	0.400

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	23.2	104.4	72.2	102.6	72.5	100.8	72.9	81.6	89.9	80.8	89.0	80.0	88.1	8.5	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.9	156.1	84.2	79.0	151.3	83.5	78.3	147.9	83.0	45.6	89.7	44.6	89.1	43.9	88.4	98.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	18.1	16.1	14.3	47.4	107.8	2.3	2.2	45.8	109.4	44.5	110.6	43.2	111.9	41.6	113.4	
(5)				42.5	86.3	1.8	5.2	41.2	90.0	40.1	93.0	39.1	95.9	37.8	99.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.0	86.0	85.2	85.3	83.0	77.7	72.9	72.6	74.9	82.5	87.2	89.1	87.7
	DBStd	6.85	3.24	3.25	2.92	3.42	4.60	4.87	4.37	4.36	3.92	3.15	2.88	3.66
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	3	0	0	0	0	0	2	1	0	0	0	0	0
	CDD50	11674	1117	986	1093	991	858	687	700	773	974	1153	1172	1170
	CDD65	6203	652	566	628	541	393	239	236	309	524	688	722	705
	CDH74	77897	8024	6661	7571	6566	4411	2558	2779	3975	6832	9349	9938	9233
	CDH80	41855	4068	3227	3863	3569	2139	1043	1172	1990	4064	5648	5948	5124
Wind	WSAvg	6.2	6.3	5.8	5.4	5.3	6.1	6.6	6.2	6.2	6.4	6.9	6.9	6.6
Precipitation	PrecAvg	30.7	7.3	8.3	5.4	1.1	0.2	0.0	0.1	0.0	0.1	0.8	2.7	5.1
	PrecMax	48.0	13.9	15.9	18.0	5.2	2.6	0.6	1.2	0.0	1.1	3.2	9.8	13.1
	PrecMin	20.6	1.6	2.3	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2
	PrecStd	7.5	3.6	3.7	4.4	1.5	0.5	0.1	0.3	0.0	0.2	0.9	2.2	3.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	103.5	102.6	102.0	100.2	96.8	93.6	93.6	96.9	101.6	105.5	106.7	107.2
		MCWB	74.9	76.1	74.2	72.4	69.9	69.3	66.6	67.8	68.8	70.0	70.9	73.1
	2%	DB	100.4	100.0	99.2	98.3	94.7	91.3	91.5	94.4	99.9	103.4	104.4	103.7
		MCWB	76.2	76.7	74.9	71.8	68.2	67.8	65.1	66.4	68.1	70.0	71.8	74.5
	5%	DB	97.9	97.3	97.2	96.6	92.9	89.1	89.4	92.3	98.3	101.6	102.4	100.6
		MCWB	77.3	77.5	74.5	71.3	67.6	65.5	63.7	64.4	67.9	70.1	72.4	76.1
	10%	DB	95.2	94.6	95.1	94.7	90.6	86.4	87.0	90.0	96.5	99.6	100.3	97.7
		MCWB	77.9	78.0	74.7	70.4	66.7	63.4	62.5	63.0	67.5	70.4	73.4	76.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	82.2	82.7	82.7	79.7	79.4	74.2	71.5	72.6	76.5	79.2	80.8	81.8
		MCDB	91.5	91.2	91.1	87.8	84.8	84.8	86.8	87.8	86.0	88.9	88.7	89.9
	2%	WB	81.1	81.1	80.9	78.1	76.2	71.2	69.4	70.7	74.8	77.8	79.4	80.7
		MCDB	89.6	89.4	88.8	86.0	83.6	84.8	84.6	86.4	85.8	87.5	88.1	89.1
	5%	WB	80.4	80.3	79.7	77.2	73.7	69.0	67.7	69.0	73.5	76.8	78.6	79.9
		MCDB	88.6	88.4	87.3	85.1	84.2	83.0	81.8	84.3	86.0	86.8	87.7	88.4
	10%	WB	79.8	79.6	78.9	76.2	71.5	66.9	65.4	67.3	72.3	75.8	77.8	79.2
		MCDB	87.8	87.3	86.1	84.5	83.1	81.4	80.4	82.4	86.4	86.8	87.6	87.7
Mean Daily Temperature Range	5% DB	MDBR	17.6	17.3	18.8	22.8	24.6	25.4	27.8	30.0	28.8	25.8	23.2	19.6
		MCDBR	21.9	22.4	23.0	25.5	26.7	27.5	30.8	31.8	29.7	28.5	26.4	24.3
		MCWBR	6.5	6.7	6.0	6.3	7.1	8.0	9.3	9.0	8.0	7.9	7.3	6.5
	5% WB	MCDBR	18.2	18.2	18.0	19.2	20.3	25.1	26.4	27.9	26.6	24.3	22.6	20.4
		MCWBR	6.1	6.4	5.6	5.7	7.2	8.3	9.6	8.3	7.5	7.3	6.8	6.2
Clear-Sky Solar Irradiance	taub		0.415	0.413	0.399	0.379	0.370	0.345	0.347	0.362	0.425	0.447	0.456	0.427
	taud		2.466	2.473	2.508	2.505	2.468	2.497	2.471	2.406	2.278	2.255	2.283	2.415
	Ebn at Noon		295	293	291	285	277	279	281	286	279	280	281	291
	Edh at Noon		38	37	35	33	32	30	31	36	43	46	45	40
All-Sky Solar Radiation	RadAvg		1981	1967	1918	1892	1706	1610	1706	1950	2153	2243	2210	2043
	RadStd		157	240	219	124	111	70	59	55	87	118	92	129